

Lectures on

# Conformal mappings



Delivered By

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## 8.1 Transformations or Mappings

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The set of equations

$$\begin{aligned}u &= u(x, y) \\ v &= v(x, y)\end{aligned}\tag{8.1}$$

defines, in general, a *transformation* or *mapping*, which establishes a correspondence between points in the  $uv$  and  $xy$  planes. The equations (8.1) are called *transformation equations*. If to each point of the  $uv$  plane, there corresponds one and only one point of the  $xy$  plane, and conversely, we speak of a *one-to-one* transformation or mapping. In such a case, a set of points in the  $xy$  plane (such as a curve or region) is *mapped* into a set of points in the  $uv$  plane (curve or region) and conversely. The corresponding sets of points in the two planes are often called *images* of each other.

## 8.11 Mapping of a Half Plane onto a Circle

Let  $z_0$  be any point  $P$  in the upper half of the  $z$  plane denoted by  $\mathcal{R}$  in Fig. 8-5. Then, the transformation

$$w = e^{j\theta_0} \left( \frac{z - z_0}{z - \bar{z}_0} \right) \quad (8.8)$$

maps this upper half plane in a one-to-one manner onto the interior  $\mathcal{R}'$  of the unit circle  $|w| = 1$ . Each point of the  $x$  axis is mapped to the boundary of the circle. The constant  $\theta_0$  can be determined by making one particular point of the  $x$  axis correspond to a given point on the circle.

In the above figures, we have used the convention that unprimed points such as  $A, B, C$ , etc., in the  $z$  plane correspond to primed points  $A', B', C'$ , etc., in the  $w$  plane. Also, in the case where points are at infinity, we indicate this by an arrow such as at  $A$  and  $F$  in Fig. 8-5, which correspond, respectively, to  $A'$  and  $F'$  (the same point) in Fig. 8-6. As point  $z$  moves on the boundary of  $\mathcal{R}$  [i.e., the real axis] from  $-\infty$  (point  $A$ ) to  $+\infty$  (point  $F$ ),  $w$  moves counterclockwise along the unit circle from  $A'$  back to  $A'$ .

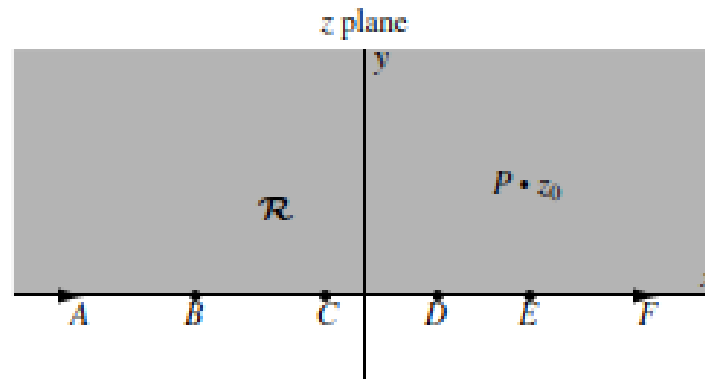


Fig. 8-5

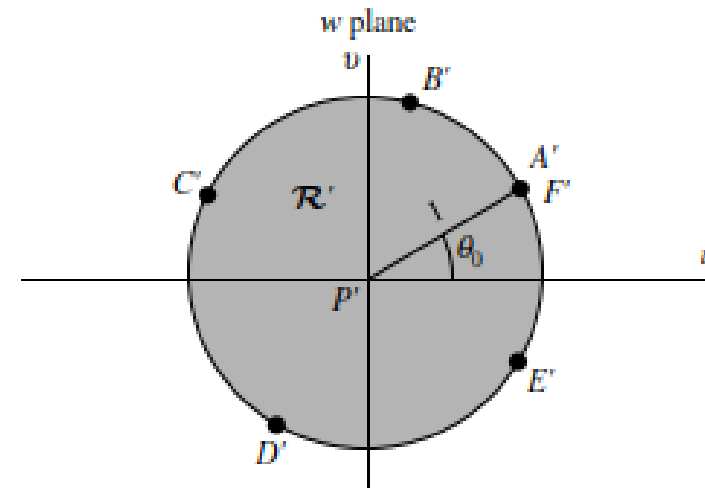


Fig. 8-6

## Some General Transformations

In the following,  $\alpha, \beta$  are given complex constants while  $a, \theta_0$  are real constants.

1. **Translation.**  $w = z + \beta$

By this transformation, figures in the  $z$  plane are *displaced* or *translated* in the direction of vector  $\beta$ .

2. **Rotation.**  $w = e^{i\theta_0}z$

By this transformation, figures in the  $z$  plane are rotated through an angle  $\theta_0$ . If  $\theta_0 > 0$ , the rotation is counterclockwise while, if  $\theta_0 < 0$ , the rotation is clockwise.

3. **Stretching.**  $w = az$

By this transformation, figures in the  $z$  plane are stretched (or contracted) in the direction  $z$  if  $a > 1$  (or  $0 < a < 1$ ). We consider contraction as a special case of stretching.

4. **Inversion.**  $w = 1/z$

## 8.9 The Linear Transformation

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The transformation

$$w = \alpha z + \beta \quad (8.5)$$

where  $\alpha$  and  $\beta$  are given complex constants, is called a *linear transformation*. Letting  $\alpha = ae^{i\theta_0}$ , we see that a general linear transformation is a combination of the transformations of translation, rotation, and stretching.

## 8.10 The Bilinear or Fractional Transformation

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The transformation

$$w = \frac{\alpha z + \beta}{\gamma z + \delta}, \quad \alpha\delta - \beta\gamma \neq 0 \quad (8.6)$$

is called a *bilinear or fractional transformation*. This transformation can be considered as a combination of the transformations of translation, rotation, stretching, and inversion.

## ✓ 8.6 Fixed or Invariant Points of a Transformation

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Suppose that we superimpose the  $w$  plane on the  $z$  plane so that the coordinate axes coincide and there is essentially only one plane. Then we can think of the transformation  $w = f(z)$  as taking certain points of the plane into other points. Points for which  $z = f(z)$  are called the *fixed* or *invariant points* of the transformation.

**EXAMPLE 8.1:** The fixed or invariant points of the transformation  $w = z^2$  are solutions of  $z^2 = z$ , i.e.,  $z = 0, 1$ .

# Some Special Mappings

See page 248 (259)

## 8.4 Conformal Mapping

Suppose that under transformation (8.1), point  $(x_0, y_0)$  of the  $xy$  plane is mapped into point  $(u_0, v_0)$  of the  $uv$  plane (Figs. 8-1 and 8-2) while curves  $C_1$  and  $C_2$  [intersecting at  $(x_0, y_0)$ ] are mapped, respectively, into curves  $C'_1$  and  $C'_2$  [intersecting at  $(u_0, v_0)$ ]. Then, if the transformation is such that the angle at  $(x_0, y_0)$  between  $C_1$  and  $C_2$  is equal to the angle at  $(u_0, v_0)$  between  $C'_1$  and  $C'_2$  both in magnitude and sense, the transformation or mapping is said to be *conformal* at  $(x_0, y_0)$ . A mapping that preserves the magnitudes of angles but not necessarily the sense is called *isogonal*.

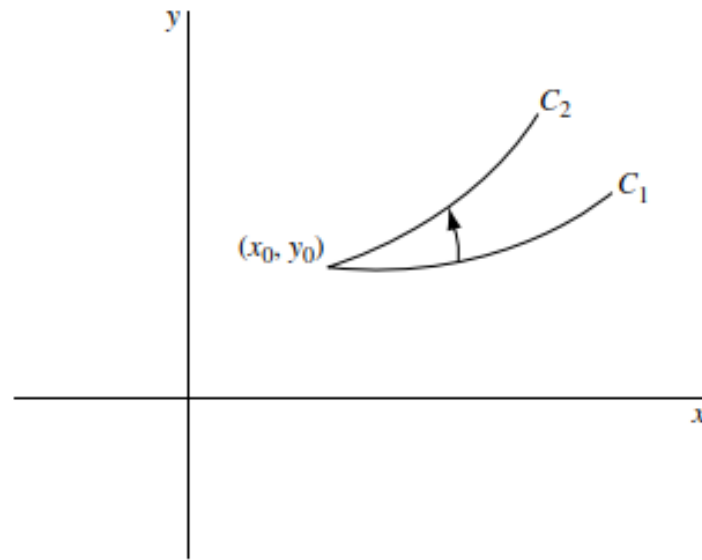


Fig. 8-1

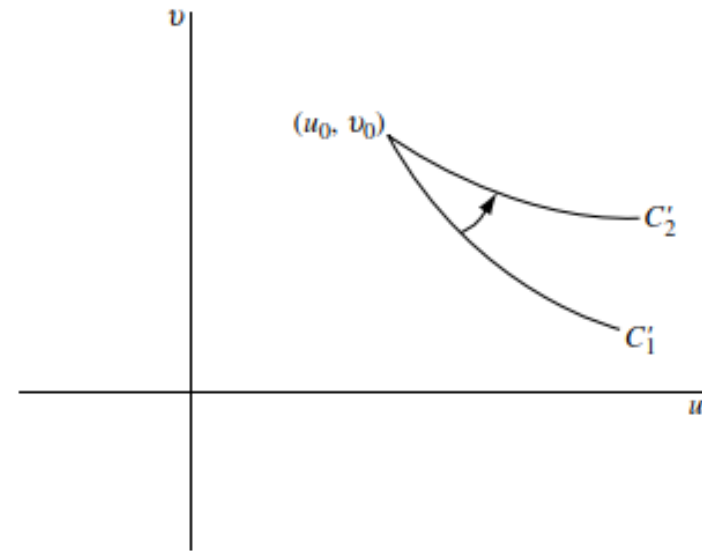


Fig. 8-2



## Transformations

**8.1.** Let the rectangular region  $\mathcal{R}$  [Fig. 8-61] in the  $z$  plane be bounded by  $x = 0$ ,  $y = 0$ ,  $x = 2$ ,  $y = 1$ . Determine the region  $\mathcal{R}'$  of the  $w$  plane into which  $\mathcal{R}$  is mapped under the transformations:

(a)  $w = z + (1 - 2i)$ , (b)  $w = \sqrt{2}e^{i\pi/4}z$ , (c)  $w = \sqrt{2}e^{i\pi/4}z + (1 - 2i)$ .

(a) Given  $w = z + (1 - 2i)$ . Then  $u + iv = x + iy + 1 - 2i = (x + 1) + i(y - 2)$  and  $u = x + 1$ ,  $v = y - 2$ .

Line  $x = 0$  is mapped into  $u = 1$ ;  $y = 0$  into  $v = -2$ ;  $x = 2$  into  $u = 3$ ;  $y = 1$  into  $v = -1$  [Fig. 8-62].

Similarly, we can show that each point of  $\mathcal{R}$  is mapped into one and only one point of  $\mathcal{R}'$  and conversely.

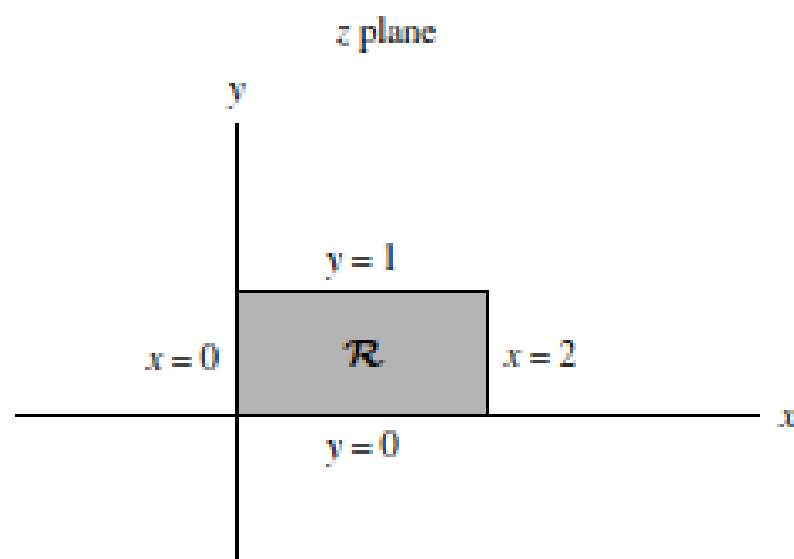


Fig. 8-61

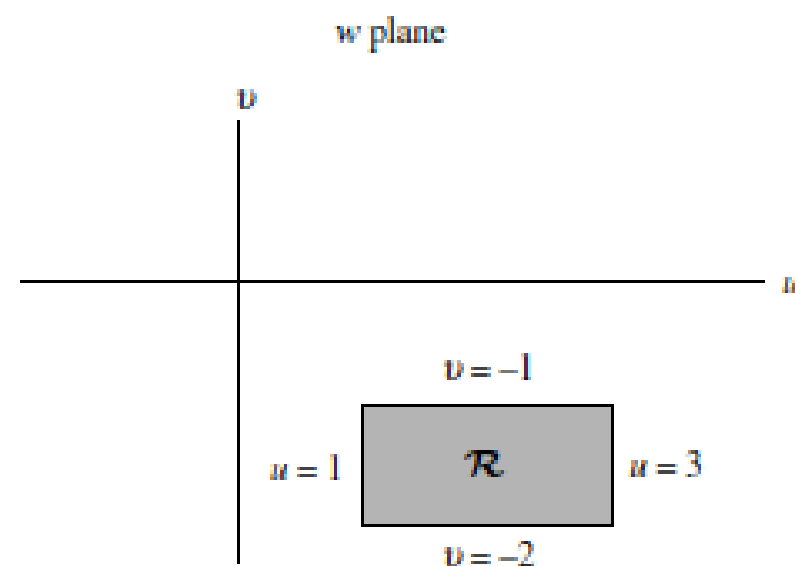


Fig. 8-62

The transformation or mapping accomplishes a *translation* of the rectangle. In general,  $w = z + \beta$  accomplishes a translation of any region.

(b) Given  $w = \sqrt{2}e^{i\pi/4}z$ . Then  $u + iv = (1 + i)(x + iy) = x - y + i(x + y)$  and  $u = x - y$ ,  $v = x + y$ .

Line  $x = 0$  is mapped into  $u = -y$ ,  $v = y$  or  $u = -v$ ;  $y = 0$  into  $u = x$ ,  $v = x$  or  $u = v$ ;  $x = 2$  into  $u = 2 - y$ ,  $v = 2 + y$  or  $u + v = 4$ ;  $y = 1$  into  $u = x - 1$ ,  $v = x + 1$  or  $v - u = 2$  [Fig. 8-64].

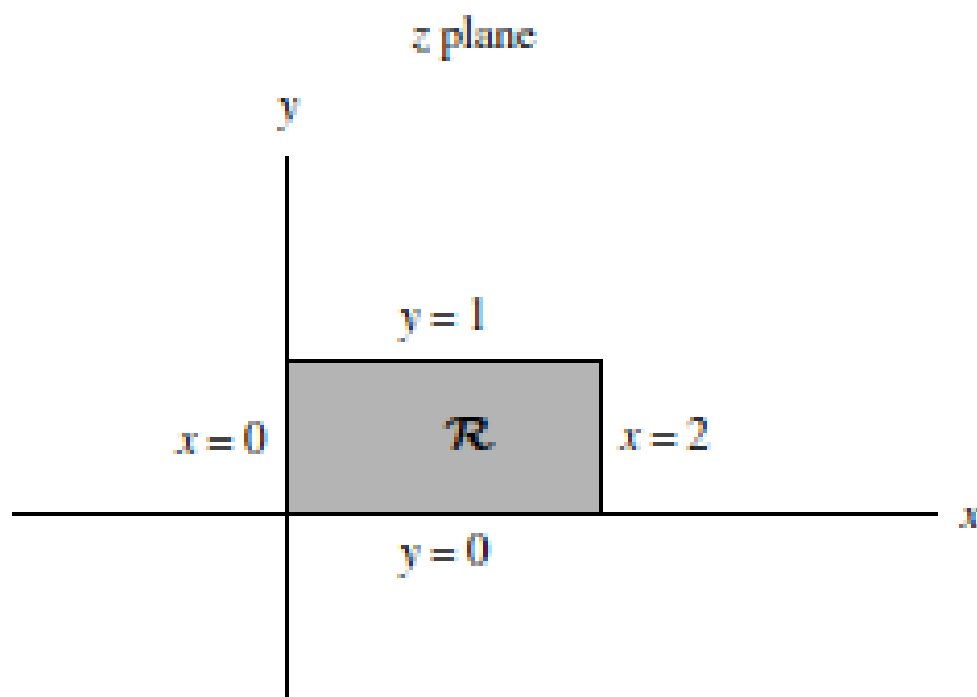


Fig. 8-63

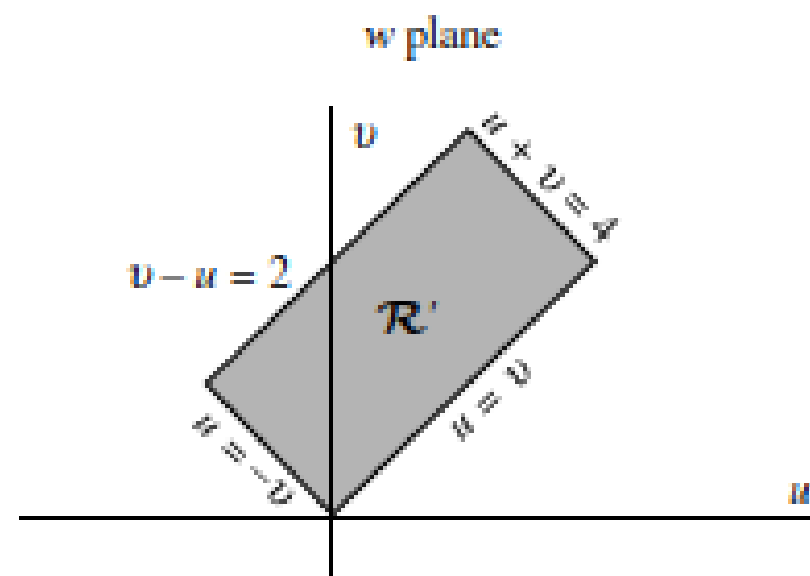


Fig. 8-64

The mapping accomplishes a *rotation* of  $\mathcal{R}$  (through angle  $\pi/4$  or  $45^\circ$ ) and a *stretching* of lengths (of magnitude  $\sqrt{2}$ ). In general, the transformation  $w = az$  accomplishes a rotation and stretching of a region.

(c) Given  $w = \sqrt{2}e^{\pi i/4}z + (1 - 2i)$ . Then  $u + iv = (1 + i)(x + iy) + 1 - 2i$  and  $u = x - y + 1$ ,  $v = x + y - 2$ .

The lines  $x = 0$ ,  $y = 0$ ,  $x = 2$ ,  $y = 1$  are mapped, respectively, into  $u + v = -1$ ,  $u - v = 3$ ,  $u + v = 3$ ,  $u - v = 1$  [Fig. 8-66].

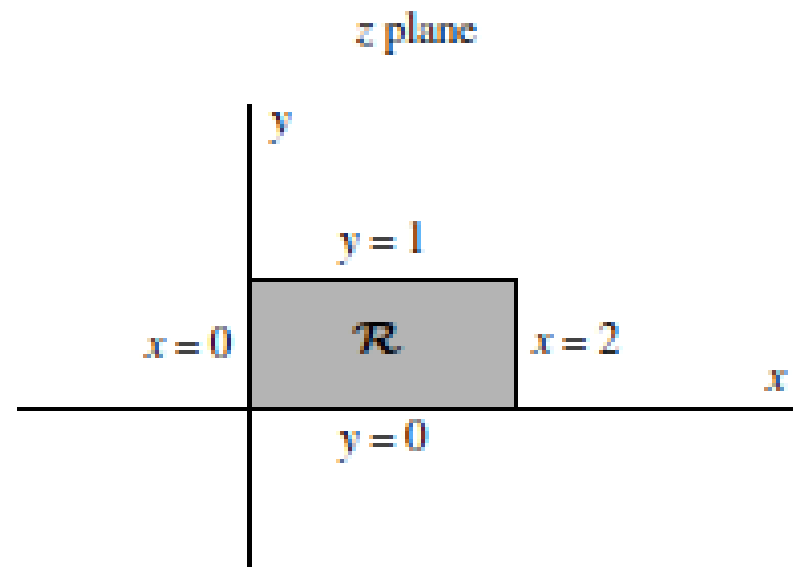


Fig. 8-65

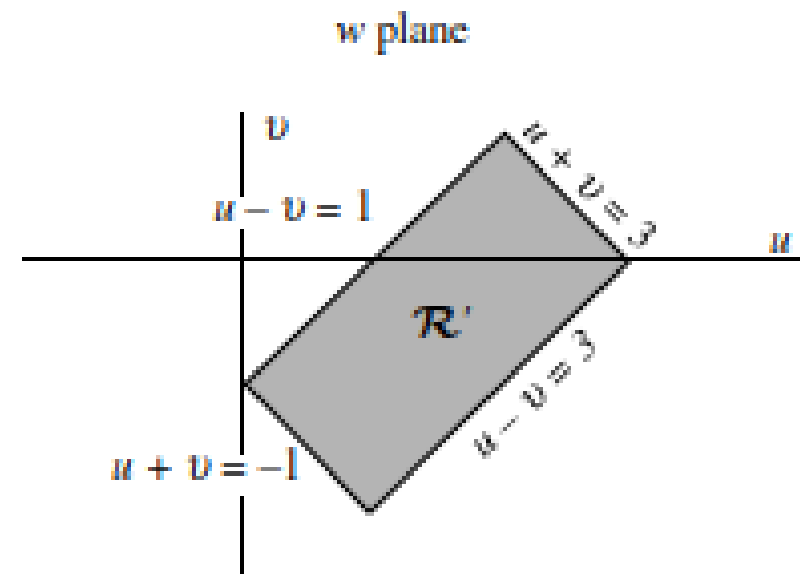


Fig. 8-66

The mapping accomplishes a rotation and stretching as in (b) and a subsequent translation. In general, the transformation  $w = \alpha z + \beta$  accomplishes a rotation, stretching, and translation. This can be considered as two successive mappings  $w = \alpha z_1$  (rotation and stretching) and  $z_1 = z + \beta/\alpha$  (translation).

**8.2.** Determine the region of the  $w$  plane into which each of the following is mapped by the transformation  $w = z^2$ . (a) First quadrant of the  $z$  plane. (b) Region bounded by  $x = 1$ ,  $y = 1$ , and  $x + y = 1$ .

**Solution**

(a) Let  $z = re^{i\theta}$ ,  $w = \rho e^{i\phi}$ . Then if  $w = z^2$ ,  $\rho e^{i\phi} = r^2 e^{2i\theta}$  and  $\rho = r^2$ ,  $\phi = 2\theta$ . Thus points in the  $z$  plane at  $(r, \theta)$  are rotated through angle  $2\theta$ . Since all points in the first quadrant [Fig. 8-67] of the  $z$  plane occupy the region  $0 \leq \theta \leq \pi/2$ , they map into  $0 \leq \phi \leq \pi$  or the upper half of the  $w$  plane [Fig. 8-68].

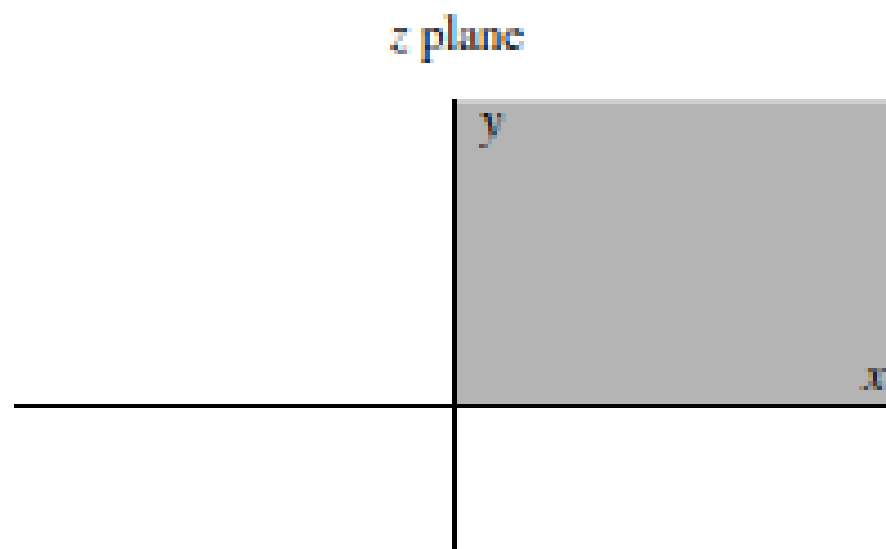


Fig. 8-67

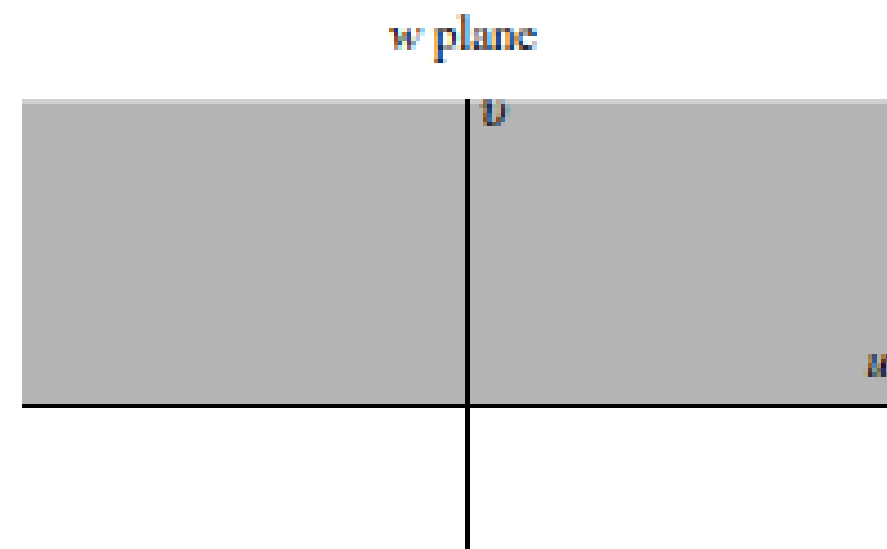


Fig. 8-68

(b) Since  $w = z^2$  is equivalent to  $u + iv = (x + iy)^2 = x^2 - y^2 + 2ixy$ , we see that  $u = x^2 - y^2$ ,  $v = 2xy$ . Then line  $x = 1$  maps into  $u = 1 - y^2$ ,  $v = 2y$  or  $u = 1 - v^2/4$ ; line  $y = 1$  into  $u = x^2 - 1$ ,  $v = 2x$  or  $u = v^2/4 - 1$ ; line  $x + y = 1$  or  $y = 1 - x$  into  $u = x^2 - (1 - x)^2 = 2x - 1$ ,  $v = 2x(1 - x) = 2x - 2x^2$  or  $v = \frac{1}{2}(1 - u^2)$  on eliminating  $x$ .

The regions appear shaded in Figs. 8-69 and 8-70 where points  $A, B, C$  map into  $A', B', C'$ . Note that the angles of triangle  $ABC$  are equal, respectively, to the angles of curvilinear triangle  $A'B'C'$ . This is a consequence of the fact that the mapping is *conformal*.

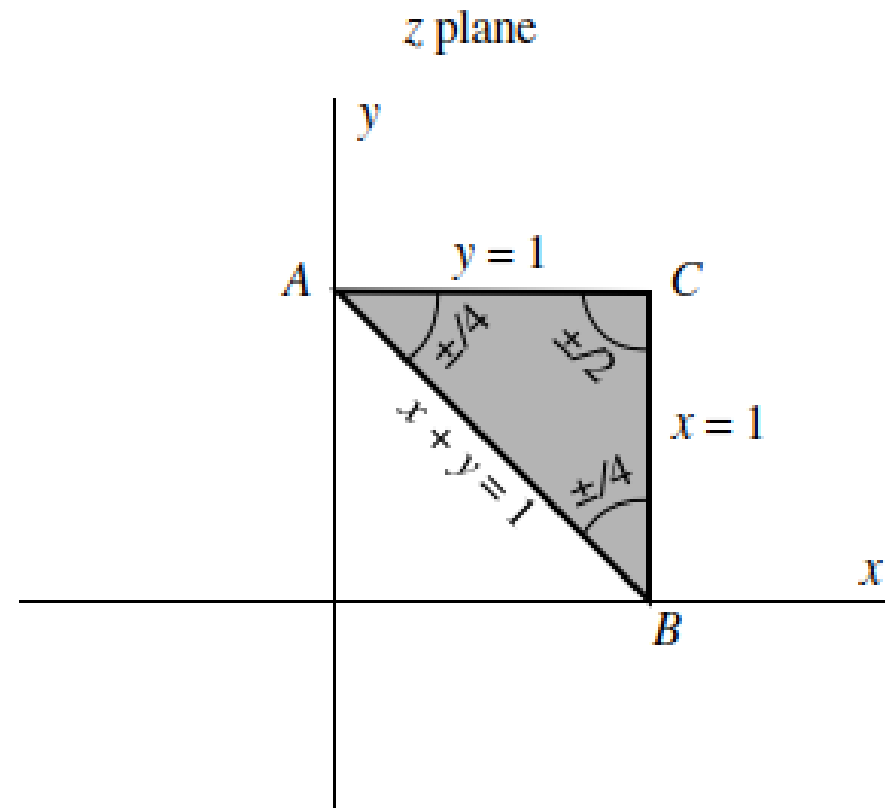


Fig. 8-69

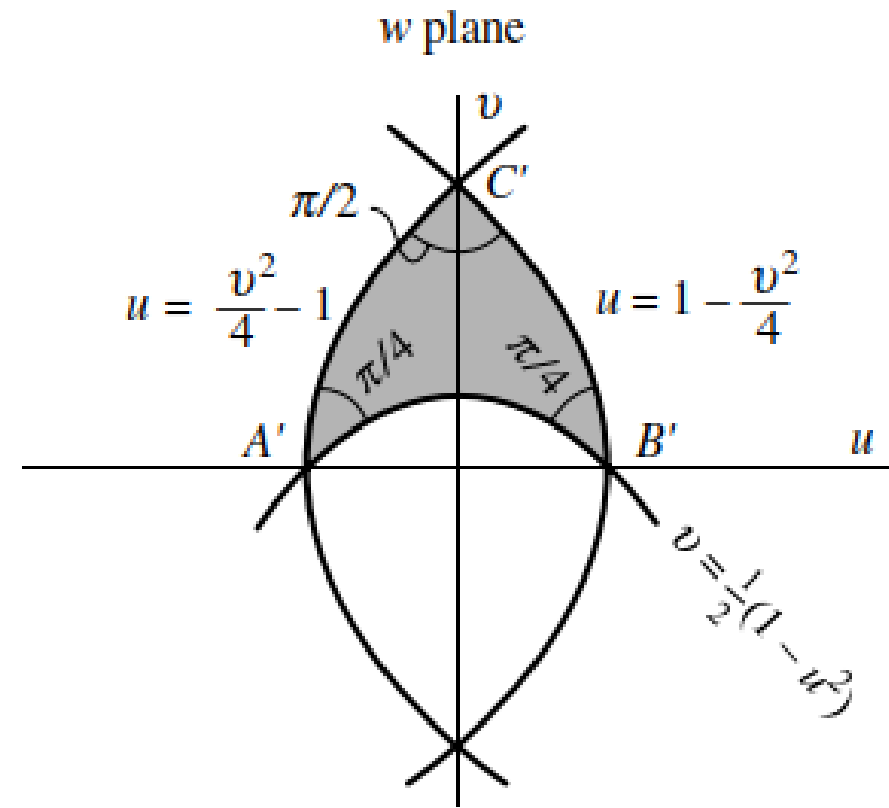


Fig. 8-70

**8.10.** Find a bilinear transformation that maps points  $z = 0, -i, -1$  into  $w = i, 1, 0$ , respectively.

**Solution**

*Method 1.* Since  $w = (\alpha z + \beta)/(\gamma z + \delta)$ , we have

$$i = \frac{\alpha(0) + \beta}{\gamma(0) + \delta} \quad (1)$$

$$1 = \frac{\alpha(-i) + \beta}{\gamma(-i) + \delta} \quad (2)$$

$$0 = \frac{\alpha(-1) + \beta}{\gamma(-1) + \delta} \quad (3)$$

From (3),  $\beta = \alpha$ . From (1),  $\delta = \beta/i = -i\alpha$ . From (2),  $\gamma = i\alpha$ . Then

$$w = \frac{\alpha z + \alpha}{i\alpha z - i\alpha} = \frac{1}{i} \left( \frac{z+1}{z-1} \right) = -i \left( \frac{z+1}{z-1} \right)$$

**8.38.** The straight lines  $y = 2x$ ,  $x + y = 6$  in the  $xy$  plane are mapped onto the  $w$  plane by means of the transformation  $w = z^2$ . (a) Show graphically the images of the straight lines in the  $w$  plane. (b) Show analytically that the angle of intersection of the straight lines is the same as the angle of intersection of their images and explain why this is so.

 **8.42.** Find a bilinear transformation that maps the upper half of the  $z$  plane into the unit circle in the  $w$  plane in such a way that  $z = i$  is mapped into  $w = 0$  while the point at infinity is mapped into  $w = -1$ .

**8.53.** Find a bilinear transformation that maps the points  $i$ ,  $-i$ ,  $1$  of the  $z$  plane into  $0$ ,  $1$ ,  $\infty$  of the  $w$  plane, respectively.

### Jacobian of a Transformation

**8.5.** Let  $w = f(z) = u + iv$  be analytic in a region  $\mathcal{R}$ . Prove that

$$\frac{\partial(u, v)}{\partial(x, y)} = |f'(z)|^2$$

#### **Solution**

If  $f(z)$  is analytic in  $\mathcal{R}$ , then the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

are satisfied in  $\mathcal{R}$ . Hence

$$\begin{aligned} \frac{\partial(u, v)}{\partial(x, y)} &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ -\frac{\partial u}{\partial y} & \frac{\partial u}{\partial x} \end{vmatrix} = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 \\ &= \left|\frac{\partial u}{\partial x} + i\frac{\partial u}{\partial y}\right|^2 = |f'(z)|^2 \end{aligned}$$

**8.6.** Find the Jacobian of the transformation in (a) Problem 8.1(c), (b) Problem 8.2, and interpret geometrically.



**8.7.** Prove that  $\frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(u, v)} = 1$ .

Thank you